Recall Proposition X from the lecture.

Proposition X. The *Euclidean inner product* obeys the properties:

- **Positivity**: $\langle \boldsymbol{v}, \boldsymbol{v} \rangle \geq 0$ for all $\boldsymbol{v} \in \mathbb{R}^n$.
- **Definiteness**: $\langle \boldsymbol{v}, \boldsymbol{v} \rangle = 0$ if and only if $\boldsymbol{v} = \mathbf{0}$.
- **Additivity in first slot**: $\langle \boldsymbol{u} + \boldsymbol{v}, \boldsymbol{w} \rangle = \langle \boldsymbol{u}, \boldsymbol{w} \rangle + \langle \boldsymbol{v}, \boldsymbol{w} \rangle$ for all $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w} \in \mathbb{R}^n$.
- **Homogeneity in first slot**: $\langle \lambda \boldsymbol{u}, \boldsymbol{v} \rangle = \lambda \langle \boldsymbol{u}, \boldsymbol{v} \rangle$ for all $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^n$ and $\lambda \in \mathbb{R}$.
- **Symmetry**: $\langle \boldsymbol{u}, \boldsymbol{v} \rangle = \langle \boldsymbol{v}, \boldsymbol{u} \rangle$ for all $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^n$.

For Questions 1–6, you are to prove the statements using only the properties of the Euclidean inner product given by Proposition X.

Question 1. Consider the properties of the Euclidean inner product given by Proposition X. Use them to show the following.

(a) Show that $\langle \boldsymbol{u}, \boldsymbol{v} + \boldsymbol{w} \rangle = \langle \boldsymbol{u}, \boldsymbol{v} \rangle + \langle \boldsymbol{u}, \boldsymbol{w} \rangle$ for all $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w} \in \mathcal{V}$.

(b) Show that $\langle \boldsymbol{u}, \lambda \boldsymbol{v} \rangle = \lambda \langle \boldsymbol{u}, \boldsymbol{v} \rangle$ for all $\boldsymbol{u}, \boldsymbol{v} \in \mathcal{V}$ and $\lambda \in \mathbb{R}$.

Question 2. Consider the properties of the Euclidean inner product (Proposition X). Let $\boldsymbol{u}$, $\boldsymbol{v}$, and $\boldsymbol{w}$ belong to $\mathbb{R}^n$. Which of the following statements is true? Justify your answer.

(a) If $\langle \boldsymbol{u}, \boldsymbol{v} \rangle = \langle \boldsymbol{u}, \boldsymbol{w} \rangle$, then $\boldsymbol{v} = \boldsymbol{w}$.

(b) If $\langle \boldsymbol{u}, \boldsymbol{v} \rangle = \langle \boldsymbol{u}, \boldsymbol{w} \rangle$ for *all* $\boldsymbol{u} \in \mathcal{V}$, then $\boldsymbol{v} = \boldsymbol{w}$.

Question 3. Using Proposition X only, prove the *parallelogram law*:

$$\|\boldsymbol{u} + \boldsymbol{v}\|^2 + \|\boldsymbol{u} - \boldsymbol{v}\|^2 = 2\|\boldsymbol{u}\|^2 + 2\|\boldsymbol{v}\|^2 \text{ for all } \boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^n.$$

Question 4. Polar identities.

(a) Using Proposition X only, show that

$$\langle \boldsymbol{u}, \boldsymbol{v} \rangle = \frac{1}{4}\left(\|\boldsymbol{u} + \boldsymbol{v}\|^2 - \|\boldsymbol{u} - \boldsymbol{v}\|^2\right) \text{ for all } \boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^n.$$

(b) (omitted from AY2022/2023) Consider the complex Euclidean inner product space. Using Proposition X only, show that

$$\langle \boldsymbol{u}, \boldsymbol{v} \rangle = \frac{1}{4}\left(\|\boldsymbol{u} + \boldsymbol{v}\|^2 - \|\boldsymbol{u} - \boldsymbol{v}\|^2 + i\|\boldsymbol{u} + i\boldsymbol{v}\|^2 - i\|\boldsymbol{u} - i\boldsymbol{v}\|^2\right) \text{ for all } \boldsymbol{u}, \boldsymbol{v} \in \mathbb{C}^n.$$

Question 5 continued on next page...

Question 5.

(a) Let let $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^n$. Use Pythogoras Theorem to show that $\|\mathrm{Proj}_{\boldsymbol{u}}(\boldsymbol{v})\|^2 \leq \|\boldsymbol{v}\|^2$.

(b) Using the formula of $\mathrm{Proj}_{\boldsymbol{u}}(\boldsymbol{v})$, prove the Cauchy-Schwartz inequality. That is, show that for all $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^n$,

$$|\langle \boldsymbol{u}, \boldsymbol{v} \rangle| \leq \|\boldsymbol{u}\| \|\boldsymbol{v}\| .$$

(c) Let $a, b \in \mathbb{R}$. Use the Cauchy-Schwartz Inequality to verify that

$$a \cos\theta + b \sin\theta \leq \sqrt{a^2 + b^2}.$$

Hint: Set $\boldsymbol{u} = (a, b)^T$ and $\boldsymbol{v} = (c, d)^T$. You need to guess c and d.

Question 6. Suppose $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_k\}$ is an orthogonal set of $\mathbb{R}^n$, and $a_1, a_2, \ldots, a_k \in \mathbb{R}$. Using Proposition X only, show that

$$\left\| \sum_{n=1}^{k} a_n \boldsymbol{v}_n \right\|^2 = \sum_{n=1}^{k} |a_n|^2 \|\boldsymbol{v}_n\|^2.$$

Question 7. Let $\boldsymbol{u}_1 = (-1, 2, 2)^T$, $\boldsymbol{u}_2 = (2, 2, -1)^T$ and $\boldsymbol{b} = (1, 0, 0)$.

(a) Show that $S = \{\boldsymbol{u}_1, \boldsymbol{u}_2\}$ is an orthogonal set.

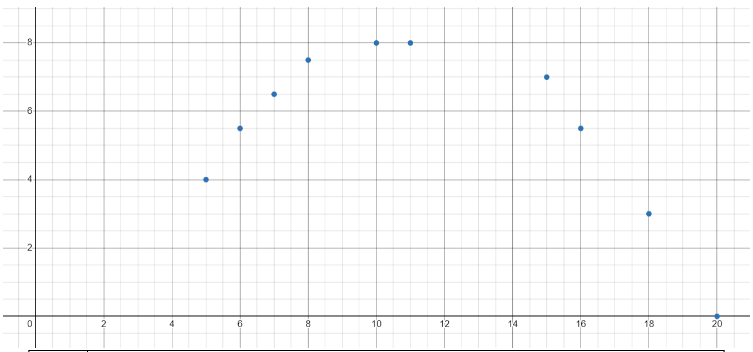
(b) Find the orthogonal projections $\boldsymbol{v}_1 \triangleq \mathrm{Proj}_{\boldsymbol{u}_1}(\boldsymbol{b})$ and $\boldsymbol{v}_2 \triangleq \mathrm{Proj}_{\boldsymbol{u}_2}(\boldsymbol{b})$.

(c) Set $\boldsymbol{v} = \boldsymbol{v}_1 + \boldsymbol{v}_2$. In what follows, we verify that $\boldsymbol{v} = \mathrm{Proj}_{\mathcal{U}}(\boldsymbol{b})$, where $\mathcal{U} = \mathrm{span}\,(\boldsymbol{u}_1, \boldsymbol{u}_2)$. Let $\boldsymbol{e} = \boldsymbol{b} - \boldsymbol{v}$. Show that $\langle \boldsymbol{e}, \boldsymbol{u}_i \rangle = 0$ for $i = 1, 2$.

(d) Let $\boldsymbol{w}_1 = (1, 0, -2)$ and $\boldsymbol{w}_2 = (0, 2, 1)$. Hence, $\mathcal{U} = \mathrm{span}\,(\boldsymbol{w}_1, \boldsymbol{w}_2)$ (you need not prove this).

Find the orthogonal projections $\boldsymbol{x}_1 \triangleq \mathrm{Proj}_{\boldsymbol{w}_1}(\boldsymbol{b})$ and $\boldsymbol{x}_2 \triangleq \mathrm{Proj}_{\boldsymbol{u}_2}(\boldsymbol{b})$. However, we observe that $\boldsymbol{v} \neq \boldsymbol{x}_1 + \boldsymbol{x}_2$. Explain $\mathrm{Proj}_{\mathcal{U}}(\boldsymbol{b})$ is not $\mathrm{Proj}_{\boldsymbol{w}_1}(\boldsymbol{b}) + \mathrm{Proj}_{\boldsymbol{u}_2}(\boldsymbol{b})$

Question 8. A gorilla threw a banana. At different (horizontal) positions x, the height h of the banana was measured (with error of course!). Let us find the parabola that best describes the banana trajectory.



i	1	2	3	4	5	6	7	8	9	10
x_i	5	6	7	8	10	11	15	16	18	20
h_i	4	5.5	6.5	7.5	8	8	7	5.5	3	0

Formally, our task is to find λ_1, λ_2, λ_3 so as to minimize the SSE $= \sum_{i=1}^{10}((\lambda_1 + \lambda_2 x_i + \lambda_3 x_i^2) - h_i)^2$.

(a) Set $\boldsymbol{x} = (\lambda_1, \lambda_2, \lambda_3^T$. Find a matrix $\boldsymbol{A}$ and a vector $\boldsymbol{b}$ so that the task is equivalent to finding $\boldsymbol{x}$ that minimizes $||\boldsymbol{Ax} - vb||^2$.

(b) Find $\boldsymbol{x}$. You can visualize your solution here! `https://www.desmos.com/calculator/zpjnl2nvqe`.